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練習題 13.7

5. 取 $A = \{(n, \frac{1}{n}) \mid n \in \mathbb{Z}^+\}$, $B = \{(n, -\frac{1}{n}) \mid n \in \mathbb{Z}^+\}$

顯然 $\forall (x, y) \notin A$, $\exists r > 0$, 使得 $B_r(x, y) \subset A^c$, 即 A^c 是開集 $\Leftrightarrow A$ 是閉集

同理 B 也是閉集

因此, $0 \leq \rho(A, B) \leq \frac{1}{n} - (-\frac{1}{n}) = \frac{2}{n} \rightarrow 0, (n \rightarrow \infty)$

即 $\rho(A, B) = 0$

6. 設 $A \subset \bar{B}_r(\vec{0})$, $\forall \vec{p} \in \{\vec{p} \in \mathbb{R}^n \mid \rho(\vec{p}, A) \leq c\}$, $\forall \vec{a} \in A$

$$\|\vec{p}\| = \|\vec{p} - \vec{a} + \vec{a}\| \leq \|\vec{p} - \vec{a}\| + \|\vec{a}\| \leq \text{diam } B$$

$$\|\vec{p}\| \leq \rho(\vec{p}, A) + \sup_{\vec{a} \in A} \|\vec{a}\| \leq c + r$$

因為 $\{\vec{p} \in \mathbb{R}^n \mid \rho(\vec{p}, A) \leq c\}$ 是有界的

考慮其補集 $\{\vec{q} \in \mathbb{R}^n \mid \rho(\vec{q}, A) > c\}$, $\forall \vec{q} \in \{\vec{q} \in \mathbb{R}^n \mid \rho(\vec{q}, A) > c\}$

記 $\text{dis}(\vec{q}) = \rho(\vec{q}, A)$

取 $r < \frac{c}{2}$, 則 $B_r(\vec{q}) \subset \{\vec{q} \in \mathbb{R}^n \mid \rho(\vec{q}, A) > c\}$

因此其補集是開集, 即 $\{\vec{p} \in \mathbb{R}^n \mid \rho(\vec{p}, A) \leq c\}$ 是閉集

即 $\{\vec{p} \in \mathbb{R}^n \mid \rho(\vec{p}, A) \leq c\}$ 是有界閉集, 即緊緻集.

問題 13.7

1. 記 $A = \{\vec{a} \in \mathbb{R}^n \mid f(\vec{a}) < 0\}$, $B = \{\vec{b} \in \mathbb{R}^n \mid f(\vec{b}) > 0\}$, $E = A \cup B$, $A \cap B = \emptyset$

~~$A \subset \{\vec{a} \in \mathbb{R}^n \mid f(\vec{a}) < 0\}$~~

若 E 是連通集, 則 $\emptyset \neq A' \cap B = \bar{A} \cap B \subset \{\vec{a} \in \mathbb{R}^n \mid f(\vec{a}) \leq 0\} \cap B = \emptyset$

或 $\emptyset \neq B' \cap A \subset \emptyset$, 矛盾

所以 E 是非連通集 ($\Rightarrow$ 非道路連通集)

練習題 13.8

1. $\forall \vec{y} \in \vec{f}(\bar{E})$, $\exists \vec{x} \in \bar{E}$, 使得 $\vec{y} = \vec{f}(\vec{x})$

同時, $\exists \{\vec{x}_k\} \subset E$, $\lim_{k \rightarrow \infty} \vec{x}_k = \vec{x}$. 記 $\vec{y}_k = \vec{f}(\vec{x}_k)$, $\{\vec{y}_k\} \subset \vec{f}(E)$

因為 $\vec{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$ 連續, 所以 $\lim_{k \rightarrow \infty} \vec{y}_k = \lim_{k \rightarrow \infty} \vec{f}(\vec{x}_k) = \vec{f}(\vec{x}) = \vec{y}$

因此 $\vec{y} \in \overline{\vec{f}(E)}$, 即 $\vec{f}(E) \subset \overline{\vec{f}(E)}$

要使 $\vec{f}(E) = \overline{\vec{f}(E)}$, 即 $\overline{\vec{f}(E)} \setminus \vec{f}(E) = \emptyset$

因此 $\forall \vec{y} \in \overline{\vec{f}(E)}$, $\exists \{\vec{y}_k\} \subset \vec{f}(E)$, 使得 $\lim_{k \rightarrow \infty} \vec{y}_k = \vec{y}$

記 $\vec{f}^{-1}(\{\vec{y}_k\}) = \{\vec{x} \mid \exists k \in \mathbb{Z}^+, \vec{y}_k = \vec{f}(\vec{x})\}$, 記 $\vec{f}^{-1}(\{\vec{y}\}) = \{\vec{x} \in E \mid \exists k \in \mathbb{Z}^+, \vec{y}_k = \vec{f}(\vec{x})\}$

若存在 $\{\vec{x}_{k_i}\} \subset \vec{f}^{-1}(\{\vec{y}_{k_i}\})$ 且 $\vec{x}_{k_i} \rightarrow \vec{x}$ ($k \rightarrow \infty$), 則 (設 $\vec{y}_{k_i} = \vec{f}(\vec{x}_{k_i})$)

$\vec{y} = \lim_{k \rightarrow \infty} \vec{f}(\vec{x}_{k_i}) = \vec{f}(\vec{x})$. 又知 $\{\vec{x}_{k_i}\} \subset E$, 所以 $\vec{x} \in \bar{E} \Rightarrow \vec{y} \in \vec{f}(\bar{E})$

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2. (1) $\forall \vec{g} \in \overline{G(\vec{f})}, \exists \{\vec{g}_k\} \subset G(\vec{f}), \vec{g}_k \rightarrow \vec{g} (k \rightarrow \infty)$

因為 $\forall \vec{g}_k \in G(\vec{f})$, 有且只有一個 $x_k \in E$, 使得 $\vec{g}_k = (x_k, \vec{f}(x_k))$

又因為 $\lim_{k \rightarrow \infty} \vec{g}_k = \vec{g}$, 所以 $\exists x \in \bar{E}$, 使得 $\lim_{k \rightarrow \infty} x_k = x$, $\vec{g} = \lim_{k \rightarrow \infty} \vec{g}_k = (x, \vec{f}(x))$

又因為 E 是閉集, 所以 $x \in \bar{E} = E \Rightarrow \vec{g} = (x, \vec{f}(x)) \in G(\vec{f})$

即有 $G(\vec{f}) \subset \overline{G(\vec{f})}$, 又知 $G(\vec{f}) \subset \overline{G(\vec{f})}$, 所以 $G(\vec{f}) = \overline{G(\vec{f})} \Leftrightarrow$

因此 $G(\vec{f})$ 是 $\mathbb{R}^{m+n}$ 中的閉集

~~(1) E 是緊緻集 $\Leftrightarrow E$ 是列緊集 $\Leftrightarrow \forall x \in E, \exists \{x_k\} \subset E$~~

(2) 從 $G(\vec{f})$ 中任取一個點列 $\{\vec{g}_k\} = \{(x_k, \vec{f}(x_k))\}$

因為 E 是緊緻集, 所以 E 是列緊集

因此, $\exists \{x_{t_k}\} \subset \{x_k\}, \lim_{k \rightarrow \infty} x_{t_k} = x \in E$

因為 $\vec{f}$ 連續, 所以 $\lim_{k \rightarrow \infty} \vec{f}(x_{t_k}) = \vec{f}(x)$

所以 $\{\vec{g}_{t_k}\} \subset \{\vec{g}_k\}, \lim_{k \rightarrow \infty} \vec{g}_{t_k} = \lim_{k \rightarrow \infty} (x_{t_k}, \vec{f}(x_{t_k})) = (x, \vec{f}(x)) \in G(\vec{f})$

因此 $G(\vec{f})$ 是列緊集 $\Leftrightarrow G(\vec{f})$ 是緊緻集

(3) 假設 $\vec{f}$ 不連續, 則 $\exists x_0 \in E$, $\exists \{x_k\} \in E$ 且 $x_k \rightarrow x_0 (k \rightarrow \infty)$
 但 $\lim_{k \rightarrow \infty} \vec{f}(x_k) \neq \vec{f}(x_0)$

即 $\{x_{t_k}\} \subset \{x_k\}$, $\exists \varepsilon > 0$, 雖然 $|x_{t_k} - x_0| < \frac{1}{k}$, 但 $\|\vec{f}(x_{t_k}) - \vec{f}(x_0)\| \geq \varepsilon$
 記 $\vec{g}_k = (x_k, \vec{f}(x_k))$

考慮點列 $\{\vec{g}_{t_k}\}$, 因為 $G(\vec{f})$ 是緊緻集, 所以 $\exists \{\vec{g}_{t_{s_k}}\} \subset \{\vec{g}_{t_k}\}$

使得 $\exists \vec{g}_0 \in G(\vec{f})$ 且 $\lim_{k \rightarrow \infty} \vec{g}_{t_{s_k}} = \vec{g}_0$

因為 $\{x_{t_{s_k}}\} \subset \{x_{t_k}\} \subset \{x_k\}$, 所以 $\lim_{k \rightarrow \infty} x_{t_{s_k}} = x_0$

所以 $\lim_{k \rightarrow \infty} \vec{f}(x_{t_{s_k}}) = \vec{f}(x_0)$, 但 $\|\vec{f}(x_{t_{s_k}}) - \vec{f}(x_0)\| \geq \varepsilon$, 因此出現矛盾

因此 $\vec{f}$ 連續。

問題 13.8

1. 假設 $\vec{f}^{-1}$ 在 D 上不連續, 則 $\exists \vec{x}_0 \in D$, $\exists \{\vec{x}_k\} \subset D$ 且 $\vec{x}_k \rightarrow \vec{x}_0 (k \rightarrow \infty)$

但 $\exists \varepsilon > 0$, $\exists \{\vec{x}_{t_k}\} \subset \{\vec{x}_k\}$, 雖然 $\|\vec{x}_{t_k} - \vec{x}_0\| < \frac{1}{k}$, 但 $\|\vec{f}^{-1}(\vec{x}_{t_k}) - \vec{f}^{-1}(\vec{x}_0)\| \geq \varepsilon$

考慮點列 $\{\vec{f}^{-1}(\vec{x}_{t_k})\} \subset E$, 因為 E 是緊緻集, 所以 $\exists \{\vec{f}^{-1}(\vec{x}_{t_{s_k}})\} \subset \{\vec{f}^{-1}(\vec{x}_{t_k})\}$

使得 $\lim_{k \rightarrow \infty} \vec{f}^{-1}(\vec{x}_{t_{s_k}}) = \vec{s}_0 \Rightarrow \lim_{k \rightarrow \infty} \vec{x}_{t_{s_k}} = \lim_{k \rightarrow \infty} \vec{f}(\vec{f}^{-1}(\vec{x}_{t_{s_k}})) = \vec{f}(\vec{s}_0)$

又因為 $\{\vec{x}_{t_{s_k}}\} \subset \{\vec{x}_{t_k}\} \subset \{\vec{x}_k\}$, 所以 $\lim_{k \rightarrow \infty} \vec{x}_{t_{s_k}} = \vec{x}_0$

$\Rightarrow \vec{s}_0 = \vec{f}^{-1}(\vec{x}_0)$, 即 $\lim_{k \rightarrow \infty} \vec{f}^{-1}(\vec{x}_{t_{s_k}}) = \vec{f}^{-1}(\vec{x}_0)$

但這與 $\|\vec{f}^{-1}(\vec{x}_{t_{s_k}}) - \vec{f}^{-1}(\vec{x}_0)\| \geq \varepsilon$ 矛盾。

因此 $\vec{f}$ 在 D 上連續

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練習題 14.1

1.

$$(1) \frac{\partial f}{\partial u}(1,1) = \lim_{t \rightarrow 0} \frac{(1 + \frac{t}{\sqrt{2}})^2 - 1^2}{t} = \lim_{t \rightarrow 0} (\frac{t}{2} + \sqrt{2}) = \sqrt{2}$$

$$(2) \text{ 記 } \varphi(t) = f(x + \frac{7}{5}t, y + \frac{2}{5}t), \quad \varphi(t) = f(1 + \frac{7}{5}t, 1 + \frac{2}{5}t) = (\frac{7}{5}t + 1)^2 - (1 + \frac{2}{5}t)^2 = -\frac{1}{5}t(\frac{7}{5}t + 2) \\ = -\frac{7}{25}t^2 + \frac{2}{5}t$$

$$\varphi'(0) = \frac{2}{5}$$

$$3. \text{ 取 } \vec{u} = (\cos \theta, \sin \theta), \text{ 記 } \varphi(t) = f(0 + t \cos \theta, 0 + t \sin \theta) = \frac{1}{2} \sin 2\theta \quad t^2, \quad \varphi(0) = 0 \\ = |t| \sin \theta \cos \theta \\ = \frac{|t|}{2} \sin 2\theta$$

$$\frac{\varphi(t) - \varphi(0)}{t} = \frac{1}{2} \cdot \frac{|t|}{t} \sin 2\theta, \quad \text{當 } \sin 2\theta \neq 0 \text{ 時}, \quad \lim_{t \rightarrow 0^+} \frac{\varphi(t) - \varphi(0)}{t} = \frac{1}{2} \sin 2\theta$$

$$\lim_{t \rightarrow 0^-} \frac{\varphi(t) - \varphi(0)}{t} = -\frac{1}{2} \sin 2\theta$$

$$\Rightarrow \lim_{t \rightarrow 0} \frac{\varphi(t) - \varphi(0)}{t} \text{ 不存在}$$

因此只有 $\sin 2\theta = 0$ 時, $(\theta = 0, \pi)$, 方向導數存在且為 0

$$(1) \frac{\partial f}{\partial x}(0,1) = \left. \frac{df(x,1)}{dx} \right|_{x=0} = \left. \frac{d}{dx} (x+1+\sqrt{x^2+1}) \right|_{x=0} = 1 + \left. \frac{2x}{2\sqrt{x^2+1}} \right|_{x=0} = 1$$

$$\frac{\partial f}{\partial y}(0,1) = \left. \frac{df(0,y)}{dy} \right|_{y=1} = \left. \frac{d}{dy} (y+|y|) \right|_{y=1} = 2.$$

$$\frac{\partial f}{\partial x}(1,2) = \left. \frac{df(x,2)}{dx} \right|_{x=1} = \left. \frac{d}{dx} (x+2+\sqrt{x^2+4}) \right|_{x=1} = 1 + \left. \frac{x}{\sqrt{x^2+4}} \right|_{x=1} = 1 + \frac{1}{\sqrt{5}}$$

$$\frac{\partial f}{\partial y}(1,2) = \left. \frac{df(1,y)}{dy} \right|_{y=2} = \left. \frac{d}{dy} (1+y+\sqrt{1+y^2}) \right|_{y=2} = 1 + \frac{2}{\sqrt{5}}$$

$$(2) \frac{\partial f}{\partial x}(x,y) = \frac{y}{1+xy}, \quad \frac{\partial f}{\partial y}(x,y) = \frac{x}{1+xy}$$

$$\frac{\partial f}{\partial x}(1,2) = \frac{2}{3}, \quad \frac{\partial f}{\partial y}(1,2) = \frac{1}{3}$$

$$(3) \frac{\partial f}{\partial x}(x,y) = e^{x+y^2} + 2xy \cos(x^2y), \quad \frac{\partial f}{\partial y}(x,y) = 2y \cdot e^{x+y^2} + x^2 \cos(x^2y)$$

$$\frac{\partial f}{\partial x}(1,1) = e^2 + 2 \cos 1, \quad \frac{\partial f}{\partial y}(1,1) = 2e^2 + \cos 1$$

$$6. (1) \frac{\partial z}{\partial x} = y + \frac{1}{y}, \quad \frac{\partial z}{\partial y} = x - \frac{x}{y^2}$$

$$(3) \frac{\partial z}{\partial x} = \frac{(y-x)^{y-1}}{y x^{y-1}}, \quad \frac{\partial z}{\partial y} = x^y \ln x$$

$$(5) \frac{\partial z}{\partial x} = \frac{-\frac{y}{x^2}}{1+(\frac{y}{x})^2} = -\frac{y}{x^2+y^2}, \quad \frac{\partial z}{\partial y} = \frac{\frac{1}{x}}{1+(\frac{y}{x})^2} = \frac{x}{x^2+y^2}$$

$$(7) \frac{\partial u}{\partial x} = e^{x(x^2+y^2+z^2)} (3x^2+y^2+z^2), \quad \frac{\partial u}{\partial y} = e^{x(x^2+y^2+z^2)} \cdot 2xy$$

$$\frac{\partial u}{\partial z} = e^{x(x^2+y^2+z^2)} \cdot 2xz$$

$$(9) \frac{\partial u}{\partial x} = yz x^{yz-1}, \quad \frac{\partial u}{\partial y} = x^{yz} \ln x^z = z x^{yz} \ln x, \quad \frac{\partial u}{\partial z} = y x^{yz} \ln x$$

$$(11) \frac{\partial z}{\partial x_i} = \frac{1}{x_1+x_2+\dots+x_n}, \quad i=1,2,\dots,n$$

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練習題 14.2

1. 證：取 $\vec{u} = (\cos \theta, \sin \theta)$

$$\text{記 } \varphi(t) = f(0 + t\cos\theta, 0 + t\sin\theta) = \frac{t^3 \cos^2\theta \sin\theta}{t^2(t^2 \cos^2\theta + \sin^2\theta)}, \quad \varphi(0) = 0$$

$$\lim_{t \rightarrow 0} \frac{\varphi(t) - \varphi(0)}{t} = \lim_{t \rightarrow 0} \frac{\cos^2\theta \sin\theta}{t^2 \cos^2\theta + \sin^2\theta} = \frac{\cos^2\theta}{\sin\theta} \quad (\theta \neq k\pi, k \in \mathbb{Z})$$

$$\text{當 } \theta = k\pi \text{ 時, } \lim_{t \rightarrow 0} \frac{\varphi(t) - \varphi(0)}{t} = 0$$

所以 f 在原點處各個方向導數存在。

但是 $f(x, y) = \frac{1}{2} (x \neq 0)$ 時，所以在原點處不連續 $\Rightarrow$ 在原點處不可微

證：取 $\vec{u} = (\cos \theta, \sin \theta)$

$$\text{記 } \varphi(t) = \sqrt{|t \cos \theta|} \quad \text{記 } \varphi(t) = f(0 + t \cos \theta, 0 + t \sin \theta), \quad \varphi(0) = 0$$
$$= |t| \sqrt{|\cos \theta \sin \theta|} = \frac{|t|}{\sqrt{2}} \sqrt{|\sin 2\theta|}$$

$$\text{取 } \theta = \frac{\pi}{4}, \text{ 得 } \varphi(t) - \varphi(0) = \frac{|t|}{\sqrt{2}}$$

假設 $f(x, y) = \sqrt{|xy|}$ 在原點處可微，則存在 λ_1, λ_2

$$\text{使得 } \varphi(t) - \varphi(0) = \lambda_1 \cdot \frac{t}{\sqrt{2}} + \lambda_2 \cdot \frac{t}{\sqrt{2}} + o(|t|)$$

$$\text{即 } \frac{|t|}{\sqrt{2}} = \lambda_1 \cdot \frac{t}{\sqrt{2}} + \lambda_2 \cdot \frac{t}{\sqrt{2}} + o(|t|) \Leftrightarrow |t| = (\lambda_1 + \lambda_2)t + o(|t|)$$

$$\Rightarrow \lambda_1 + \lambda_2 = \lim_{t \rightarrow 0} \frac{|t|}{t}, \text{ 但 } \lim_{t \rightarrow 0} \frac{|t|}{t} \text{ 不存在}$$

所以 $f(x, y) = \sqrt{|xy|}$ 在原點處不可微

3. 取 $\vec{h} = (h_1, h_2)$

$$f(x+h_1, y+h_2) - f(x, y) = (x+h_1)(y+h_2) - xy = yh_1 + xh_2 + h_1h_2$$

$$\text{因為 } \left| \frac{h_1h_2}{\sqrt{h_1^2+h_2^2}} \right| \leq \frac{1}{2}\sqrt{h_1^2+h_2^2} = \frac{1}{2}\|\vec{h}\| \rightarrow 0, \text{ 所以 } h_1h_2 = o(\|\vec{h}\|)$$

$$\begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}, \text{ 所以 } f(x, y) \text{ 在每一點皆可微。}$$

4. (3) $\frac{\partial u}{\partial x_i} = \frac{x_i}{\sqrt{x_1^2+x_2^2+\dots+x_n^2}}$, 因此 ~~du~~ $du(t_1, t_2, \dots, t_n) = \sum_{i=1}^n \frac{t_i}{\sqrt{t_1^2+t_2^2+\dots+t_n^2}} h_i$

(4) $\frac{\partial u}{\partial x_i} = i x_i^{i-1} \cos\left(\sum_{k=1}^n x_k^k\right)$, 因此 $du(x_1, x_2, \dots, x_n) = \left(\sum_{i=1}^n i x_i^{i-1}\right) \cos\left(\sum_{k=1}^n x_k^k\right)$

5. (3) $Jf(x, y, z) = \left(\cos(y-3z) + \frac{y}{\sqrt{1-x^2y^2}}, -x \sin(y-3z) + \frac{x}{\sqrt{1-x^2y^2}}, 3x \sin(y-3z) \right)$

(4) ~~$Jf(x, y, z) = \left(\frac{y}{\sqrt{1-x^2y^2}}, -x \sin(y-3z) + \frac{x}{\sqrt{1-x^2y^2}}, 3x \sin(y-3z) \right)$~~

(4) $Jf(x_1, x_2, \dots, x_n) = \frac{1}{\sqrt{x_1^2+x_2^2+\dots+x_n^2}} (x_1, x_2, \dots, x_n)$

練習題 14.3

1. (1) $Jf(x, y) = \begin{pmatrix} y^2 - 6x & 2xy \\ 3 & -10y \end{pmatrix} \Rightarrow Jf(1, -1) = \begin{pmatrix} 7 & -2 \\ 3 & 10 \end{pmatrix}$

$d\vec{f}_{(1, -1)}^{\vec{h}} = \begin{pmatrix} 7 & -2 \\ 3 & 10 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}$

(2)

$Jf(x, y, z) = \begin{pmatrix} yz^2 & xz^2 - 8y & 2xyz \\ 3y^2 & 2(3x-2)y & -y^2 \end{pmatrix} \Rightarrow Jf(1, -2, 3) = \begin{pmatrix} -18 & 25 & -12 \\ 12 & 0 & -4 \end{pmatrix}$

~~$d\vec{f}_{(1, -2, 3)}^{\vec{h}}$~~ $d\vec{f}(1, -2, 3) = \begin{pmatrix} -18 & 25 & -12 \\ 12 & 0 & -4 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix}$

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$$(3) \vec{Jf}(xy) = \begin{pmatrix} e^x \cos(xy) - y \cdot e^x \sin(xy) & -x e^x \sin(xy) \\ e^x \sin(xy) + y \cdot e^x \cos(xy) & x e^x \cos(xy) \end{pmatrix}$$

$$\vec{Jf}\left(1, \frac{\pi}{2}\right) = e^1 \begin{pmatrix} -\frac{\pi}{2} & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -\frac{\pi e}{2} & -e \\ e & 0 \end{pmatrix}$$

$$d\vec{f}\left(1, \frac{\pi}{2}\right) = \begin{pmatrix} -\frac{\pi e}{2} & -e \\ e & 0 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}$$

$$2. (1) \vec{Jf}(r, \theta) = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}$$

$$(2) \vec{Jf}(r, \theta, z) = \begin{pmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$(3) \vec{Jf}(r, \theta, \varphi) = \begin{pmatrix} \sin \theta \cos \varphi & r \cos \theta \cos \varphi & -r \sin \theta \sin \varphi \\ \sin \theta \sin \varphi & r \cos \theta \sin \varphi & r \sin \theta \cos \varphi \\ \cos \theta & -r \sin \theta & 0 \end{pmatrix}$$